

TaCoRe: Asynchronous Co-Revision of Future Touch and Actions for Dexterous Manipulation

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Abstract

Introduction

Related Work

Method

Experiments

Experiment Setup

Main Results

Ablation Studies

We conduct ablations to isolate the effect of asynchronous high-frequency updates, future tactile latents, recursive future-latent updates, patch-aware tactile encoding, and privileged teacher action supervision. The variants are summarized in Table 2.

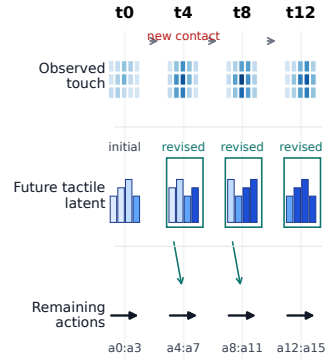
Robustness to Perturbations

We further evaluate robustness under visual and physical perturbations, including lighting changes, height variations, weight changes, and external post-grasp disturbances. Table 3 reports the average success rate for each perturbation type.

Analysis

Figure 1 visualizes how newly observed touch revises future tactile latents and the remaining action chunk during execution. Figure 2 illustrates the patch-structured tactile representation used by TaCoRe. Table 4 reports deployment latency for the main inference components.

Conclusion



New tactile evidence updates future tactile latents, which update the unexecuted action chunk.

Figure 1: Closed-loop revision of future tactile latents and actions. At each asynchronous update, newly observed touch and execution history revise the future tactile latent, which in turn updates the remaining action chunk. This draft visualization illustrates the intended analysis; final panels will be instantiated with rollout traces.

Table 1: Success rates (%) on five contact-rich dexterous manipulation tasks. Each task is evaluated over multiple rollouts, and the average is computed across tasks.

Method	Pipette Press	Orange Cob	Water Bottle	Red Button	Sponge Wipe	Avg.
ViTacFormer	—	—	—	—	—	—
RDP	—	—	—	—	—	—
Tactile-VLA	—	—	—	—	—	—
π_0	12.5	80	80	65	10	—
π_0 + tactile	31.5	0	82.5	45	25	—
$\pi_{0.5}$	29.5	0	75	45	35	—
$\pi_{0.5}$ + tactile	82	92	60	60	20	—
TaCoRe(ours)	86	100	95	70	87.5	

Table 2: Ablation study of TaCoRe. We compare the full model against five variants that remove asynchronous high-frequency updates, future tactile latents, future-latent recursive updates, patch-aware tactile encoding, or the privileged teacher action expert. Success rates (%) are reported on five contact-rich dexterous manipulation tasks and averaged across tasks.

Variant	Pipette	Cob	Bottle	Button	Wipe	Avg.
Full TaCoRe	—	—	—	—	—	—
w/o async update	—	—	—	—	—	—
w/o future tactile	—	—	—	—	—	—
w/o future-latent update	—	—	—	—	—	—
w/o Patch-aware Encoder	—	—	—	—	—	—
w/o teacher action expert	—	—	—	—	—	—

Table 3: Generalization under out-of-distribution perturbations. Each setting is evaluated on two representative tasks, and success rates (%) are averaged over the two tasks.

Setting (Tasks)	RDP	T-VLA	$\pi_{0.5}$ +Tac	TaCoRe
Visual OOD				
Bottle, Wipe	—	—	—	—
Object OOD				
Cob, Bottle	—	—	—	—
Geometric OOD				
Cob, Button	—	—	—	—
Post-grasp Pull				
Cob, Bottle	—	—	—	—
Avg.	—	—	—	—

Table 4: Inference latency of TaCoRe during deployment. Latency is reported per control cycle or asynchronous update stage.

Component	Latency (ms)
VLM cache construction	—
Patch-aware tactile encoder	—
Action expert update	—
Async correction step	—

Dense force taxels → Patch grouping → Tactile tokens

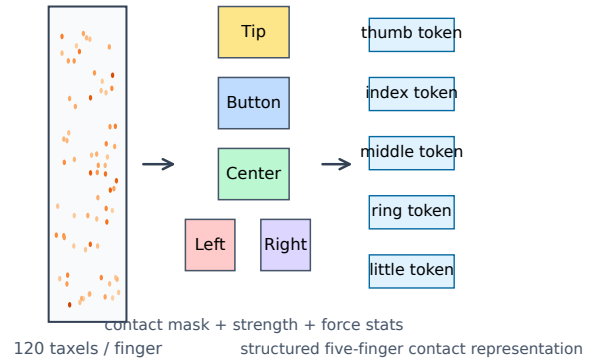


Figure 2: Patch-aware tactile representation for dexterous hands. Dense force taxels are organized into structured finger-level contact patches, producing compact tactile tokens that summarize contact mask, strength, and force statistics.

References